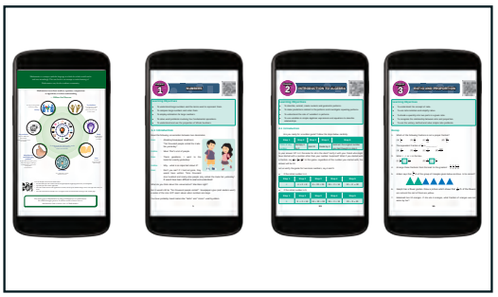
Example 11: There are four Mobile Phones in a house. At 5 a.m, all the four Mobile Phones will ring together. Thereafter, the first one rings every 15 minutes, the second one rings every 20 minutes, the third one rings every 25 minutes and the fourth one rings every 30 minutes. At what time, will the four Mobile Phones ring together again?

Solution: This is a LCM related sum. So, we need to find the LCM of 15, 20, 25 and 30.

2	15, 20, 25, 30
2	15, 10, 25, 15
5	15, 5, 25, 15
3	3, 1, 5, 3
5	1, 1, 5, 1
	1, 1, 1, 1



The LCM of 15, 20, 25 and 30 is 2 x 2 x 3 x 5 x 5
= 300 minutes = 5 x 60 minutes = 5 x 1 hour = 5 hours
Thus, the four Mobile Phones will ring together again at 10.00 a.m.

Try this

A small boy went to a town to sell a basket of wood apples. On the way, some robbers grabbed the fruits from him and ate them. The small boy went to the King and complained. The King asked him, "How many wood apples did you bring?". The boy replied, "Your Majesty! I didn't know, but I knew that if you divided my fruits into groups of 2, one fruit would be left in the basket". He continued saying that if the fruits were divided into groups of 3, 4, 5 and 6, the fruits left in the basket would be 2, 3, 4 and 5 respectively. Also, if the fruits were divided into groups of 7, no fruit would be there in the basket. Can you find the number of fruits, the small boy had initially?

(This problem is taken from the famous Mathematics problems collection book in Tamil called **"Kanakkathikaram"** under the heading of **"Wood Apple Problem"**)

1.8 Relationship between the Numbers and their HCF and LCM

Let us find the HCF and the LCM of 36 and 48. First, find the factors of 36 and 48 using division method.

36 = 2 x 2 x 3 x 3; 48 = 2 x 2 x 2 x 2 x 3
HCF = 2 x 2 x 3 = 12
LCM = 2 x 2 x 3 x 2 x 2 x 3 =144
Observe that, 36 x 48 = 144 x 12 = 1728
We find that,

2	36
2	18
3	9
3	3
	1

2	48
2	24
2	12
2	6
3	3
	1

product of two given numbers = their HCF x LCM

In general, for any 2 numbers x and y,

$$x \times y = \text{HCF}(x, y) \times \text{LCM}(x, y)$$

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